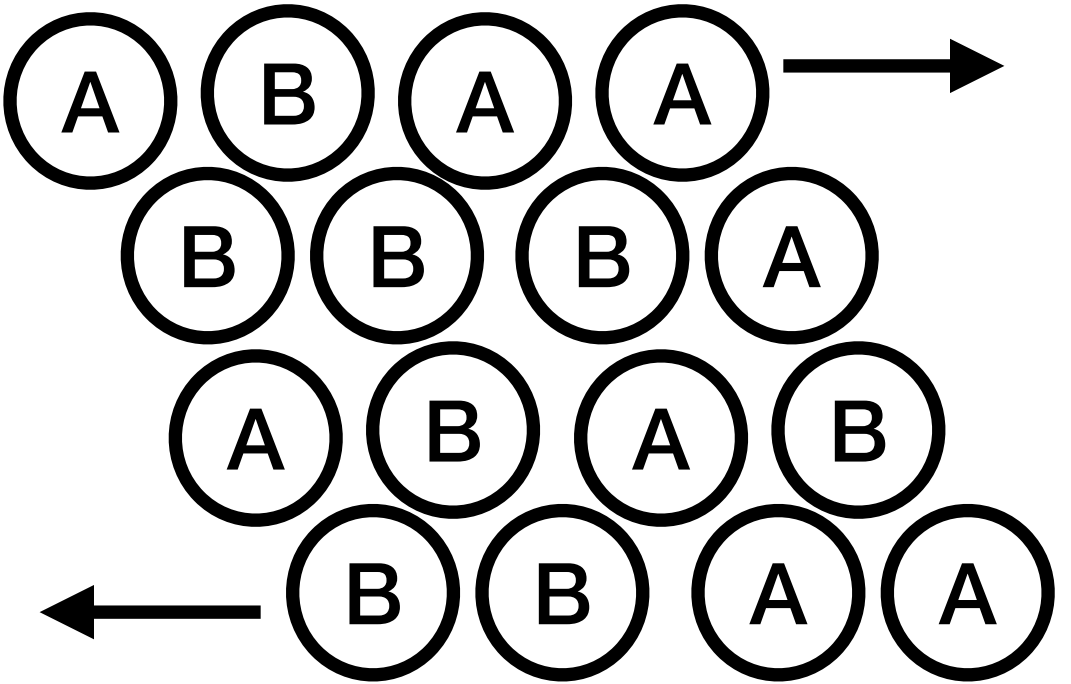
# Building a graph from a granular system

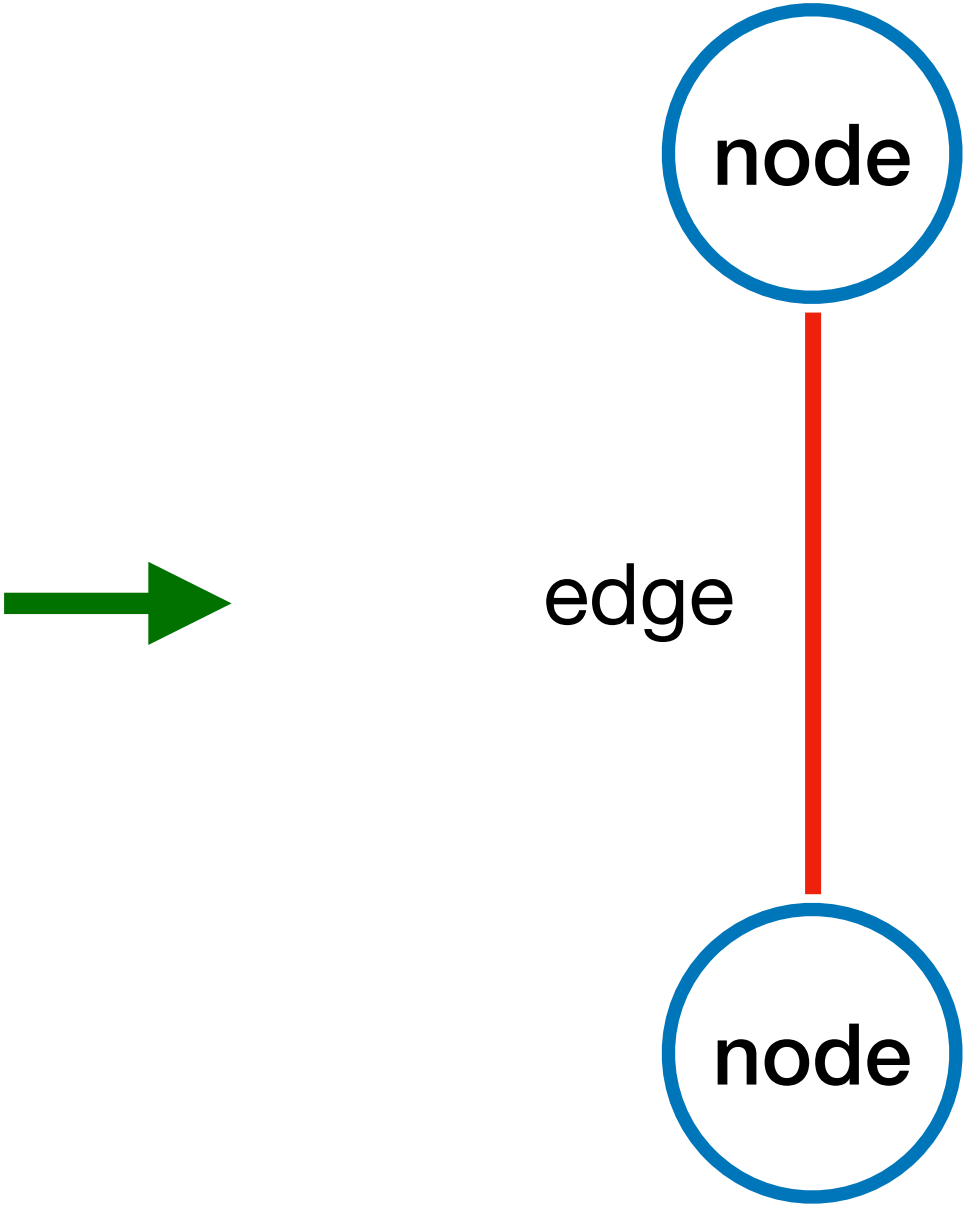
6




MD simulation of sheared granular system with binary, purely repulsive spheres

- node:
  - particle type
  - position $(x, y)$

- edge:
  - relative position vector
  - drawn between particles within distance $12\,R_B$


node
edge
node

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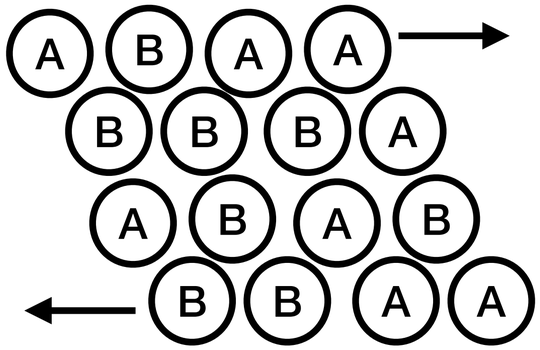


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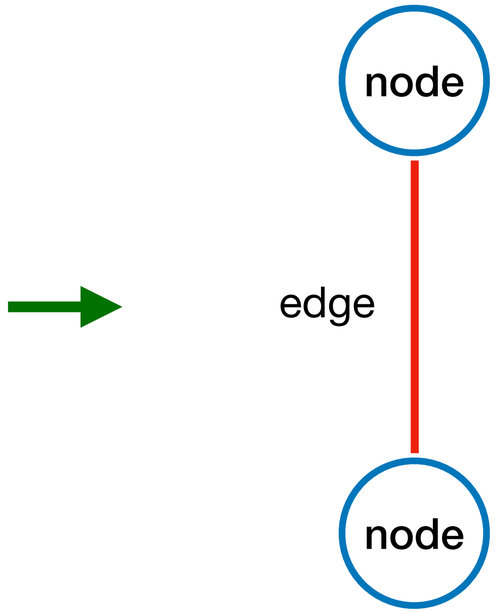


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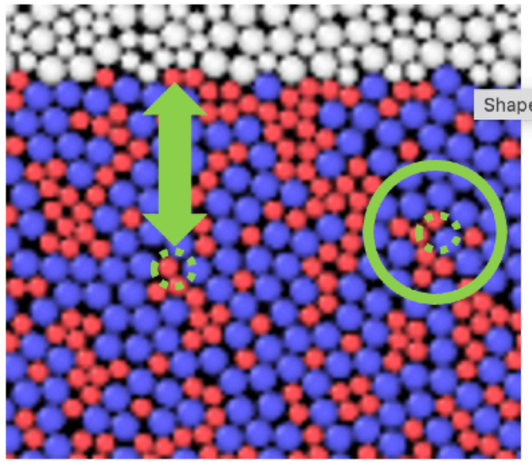
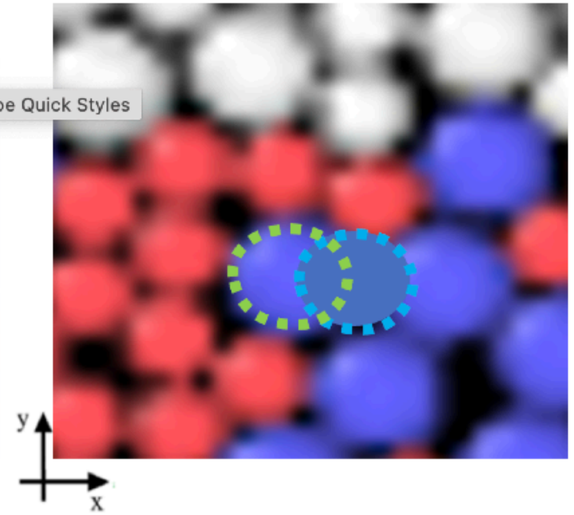
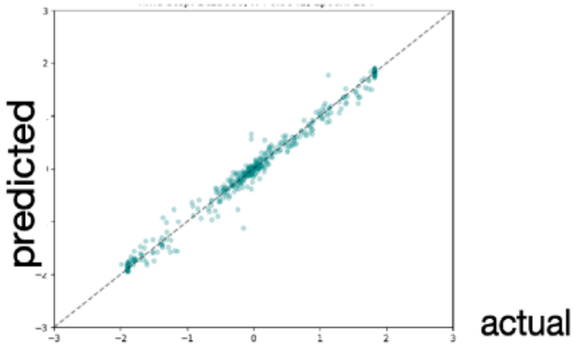
# Methods: can ML replace MD?






Toy tasks to test GNN code

- distance from wall
- number of near neighbors

Task of interest

Individual particles' displacement $\Delta x$ over time $\Delta t$

Measuring success at a task

After training, does parity plot of validation data show linear behavior with small RMSE?